Healthcare Testing @ RTX

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Question

How can we make sure that the products we develop has significant quality to go to market ?

About me

Who am i?

Name: Michael Svangren

Experience:
Masters in computer science and informatics 2012
P.hD. in computer science 2017
Post doc. 2017-19
SSI Schaefer 2019-21
RTX 2021 – present

What i do now:
Department manager for 12 people in DK
SQA manager for 5 people in DK + 5 people in RO

Area of expertise: Healthcare domain – Wireless telematry, system testing, test management, requirements engineering

What I will talk about today

RTX

- What we do
- Product examples

Test and test management

- What is quality assurance
- What is quality assurance management

Case: Healthcare – management perspective

- Intro to Healthcare domain
    - Why quality in healthcare is important - and tricky!

- Healthcare testing – Shifting left
    - The specification
    - Implementation
    - System test

RTX

What do we do?

Hardware/software development

We do Digital Enhanced Cordless Telecommunications (DECT)

We typically design for other companies -> ODM model

Systems & Solutions

Enterprise »

Wireless IP telephony products and sub-systems that include various devices and advanced cloud-based tools.

Professional Audio »

Wireless systems providing unique quality and professional-grade wireless audio devices.

Intercom »

Systems providing direct, clear, and reliable wireless communication solving challenges in many different industries.

Healthcare and monitoring »

Crucial wireless communication systems that seamlessly and reliably are embedded into a broad spectrum of high-tech medical device…

Things we excel in

Hardware
- PCB design
- Mecanics design
- PCB verification

Protocol
- RF design
- RF verification

Product integration
- High level integration of protocol and hardware

Things we outsource

Manufacturing

- PCB production at a scale

In most cases

- Large scale system testing

QA and Test Management

Is this a familiar scenario?

How to reduce software bugs - Quality Assurance

Quality Assurance (QA) is a **systematic process** used to ensure that a product or service meets specified quality standards and fulfills customer expectations.

- ISTQB foundation

<u>We ensure that the product meets the requirements and standards through testing!</u>

Internation Software Testing Qualifications Board (ISTQB)

Types of QA testing

Functional testing

Non-functional testing

Sanity Testing

Ad-Hoc Testing

Exploratory testing

Smoke Testing

Regression Testing

We need something to keep it all structured - test management

Test management *is the process of* ***organizing, planning, executing, and tracking*** *all activities involved in software testing. It ensures that the testing process is* ***systematic, efficient, and aligned with project goals****.*

- ISTQB

International Software Testing Qualifications Board (ISTQB)

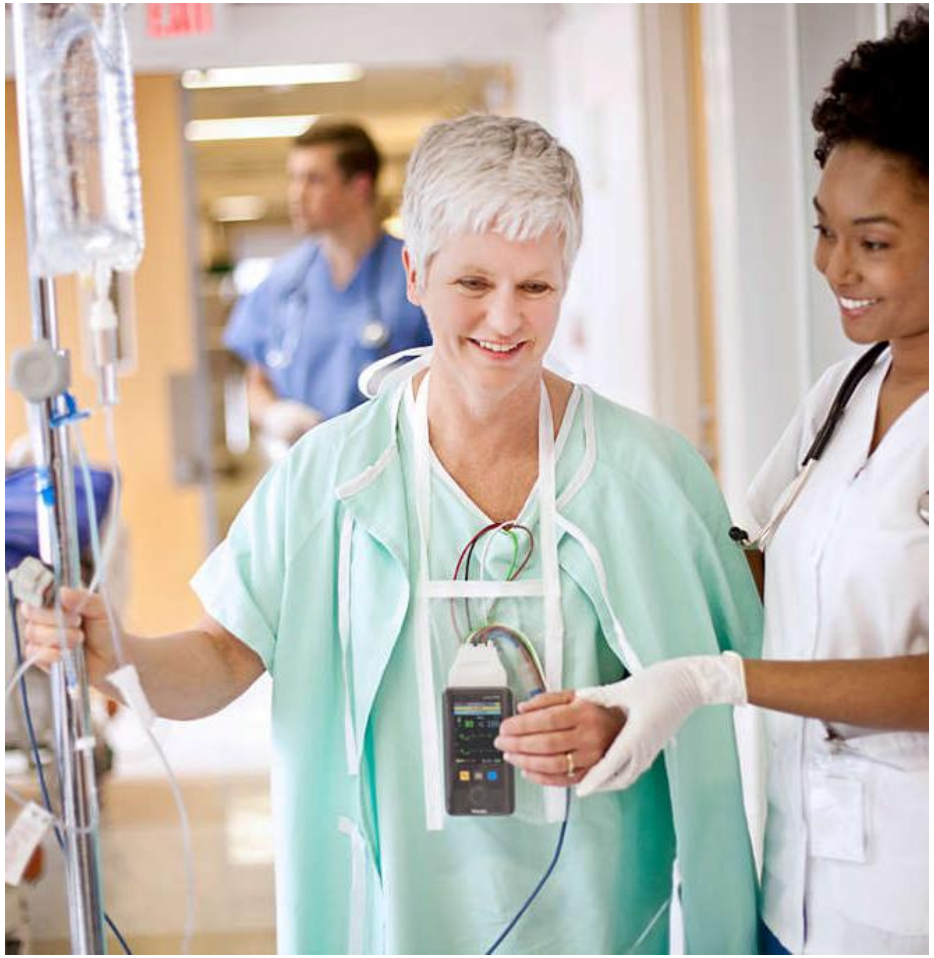
What does test management involve?

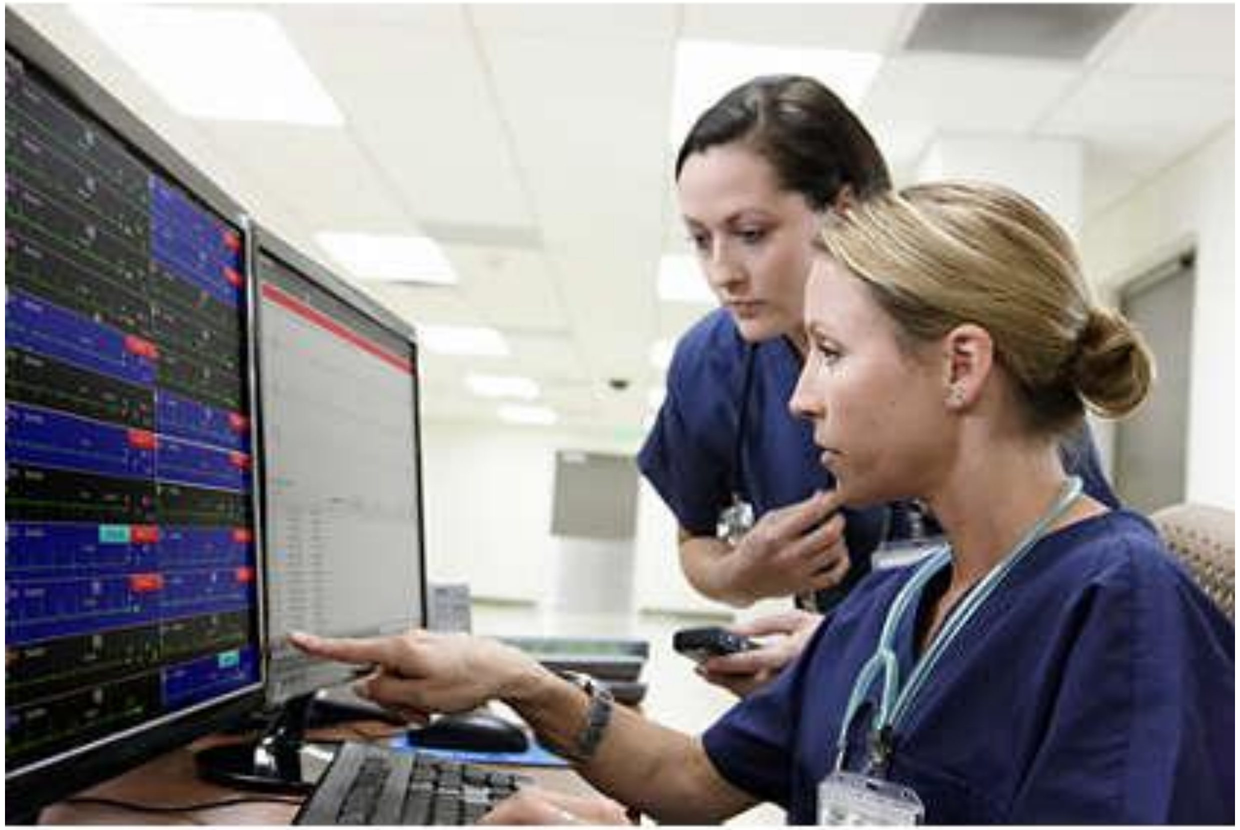
Managing all aspects of the sofware lifecycle

- Planning
- Requirement engineering
- Define and write test cases
- Execute Test Cases
- Report defects
- Track defects
- Coverage and Tracability Reports

Case: The Healthcare domain

Wireless Patient Monitoring



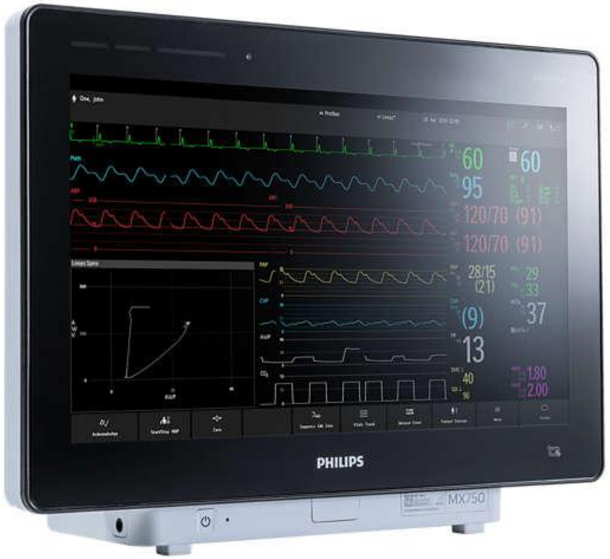


Technologies

The challenge with healthcare technology

Under strict FDA regulation (Medical Devices)
- You need processes that guide and document your requirement, development, and testing effort.

Technology must not experience any disconnects throughout its patient monitoring state
- Inconvinient for healt care personnel (they have to senc
- Potential patient loss
- Potential patient death



Healthcare – Why put effort into testing?

To gain confidence that our product is good enough to go to market (reliability, resilliance)

To prove to the customer that the product is implemented according to spec (reporting)

To prove quality, and to document that we have actually tested (To FCC, etc.)

To gain end user satisfaction (i.e., make sure that our products are reliable to customers)

Healthcare – Who is involved?

Role	Description
Sales	Sells products
Project managers	Manages project from beginning to end
Product managers	Decides what the products must do
Arcitect	Defines product functionality (requirements)
HW	Designs and implement hardware
Protocol	Design and implement low level software
PI	Integrate software and hardware
System testing	Tests the finished products and ensures that requirements are implemented correctly

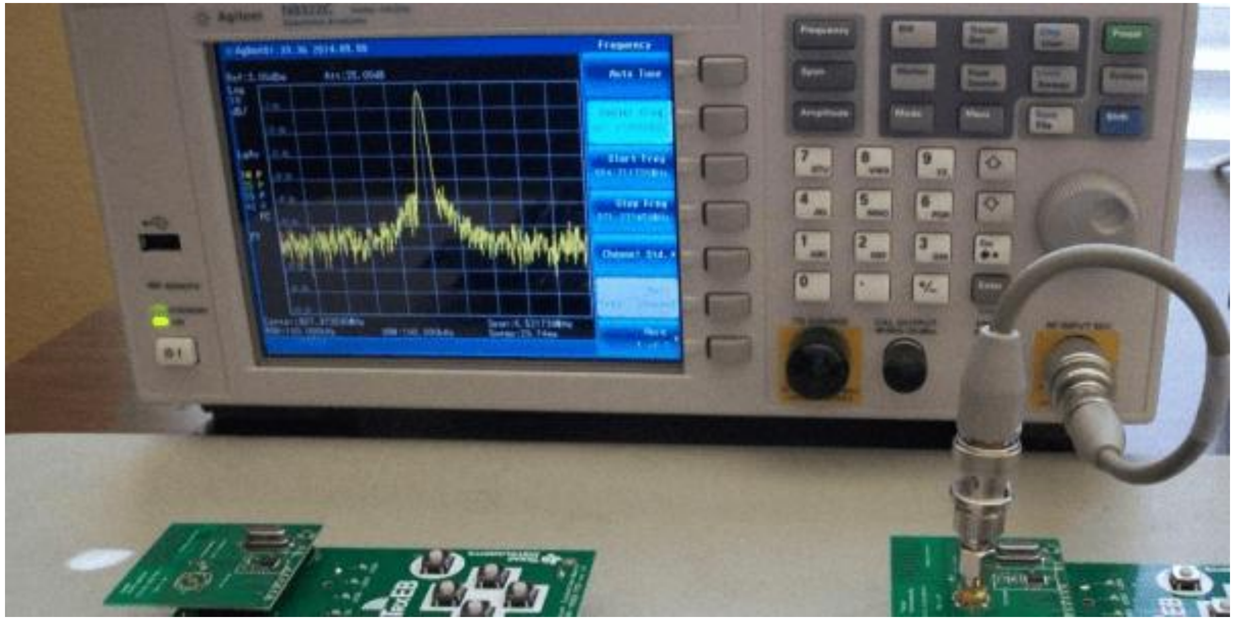
Healthcare – Test effort involved?

Role	Test types
Sales	Reviews
Project managers	Reviews, meetings
Product managers	Reviews, meetings..
Arcitect	Reviews of requirements
HW	RF testing, Climate tests, Weber tests..
Protocol	Unit testing, interface testing
PI	Unit testing, interface testing
System testing	System testing

Delimitation!

This presentation will not touch upon the details of HW/RF, PDT testing!

Although its pretty cool! (HW/RF)



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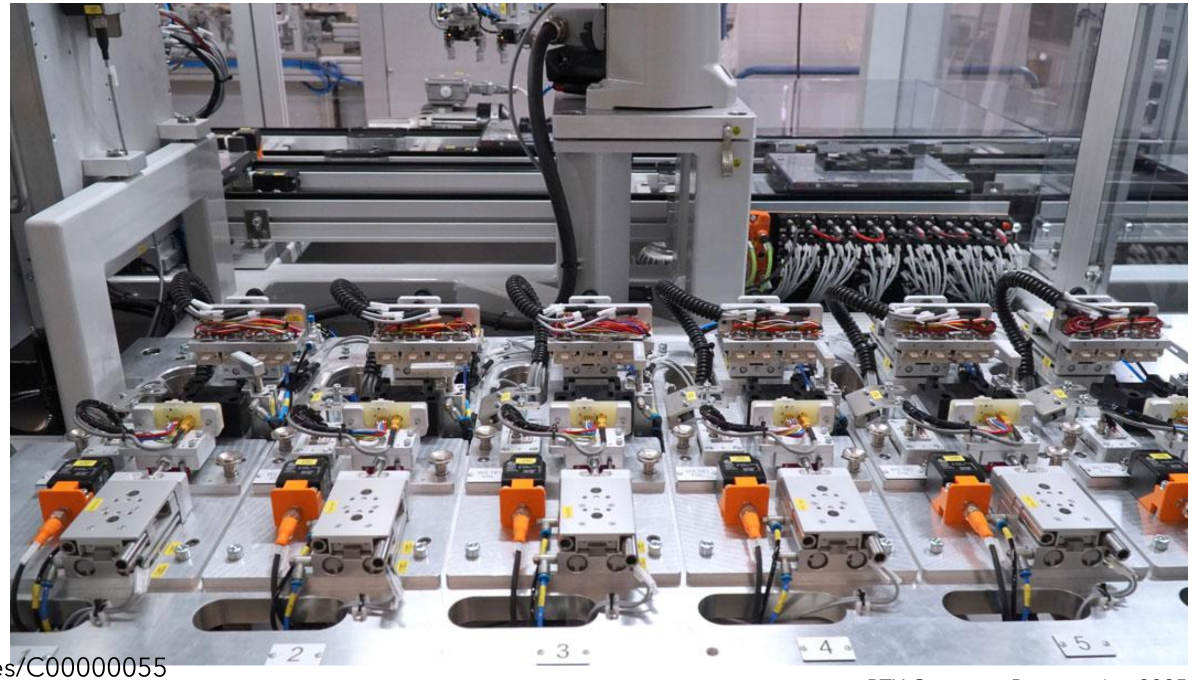
Although its pretty cool! (PDT)



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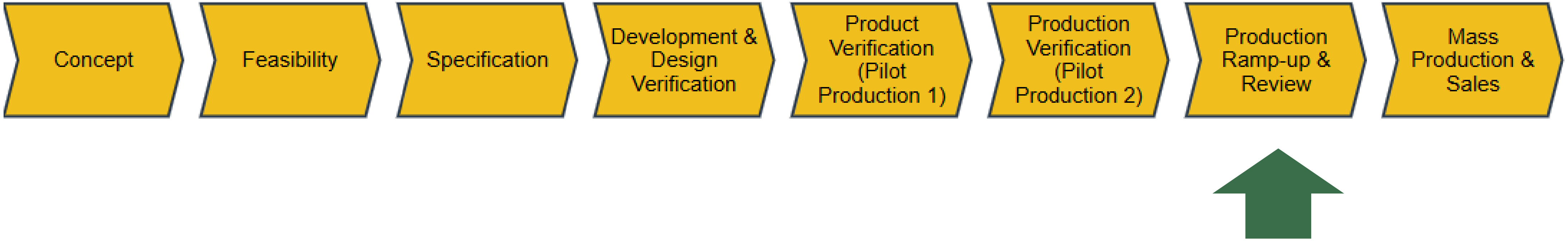
Healthcare Test Management

Healthcare – Product lifecycle

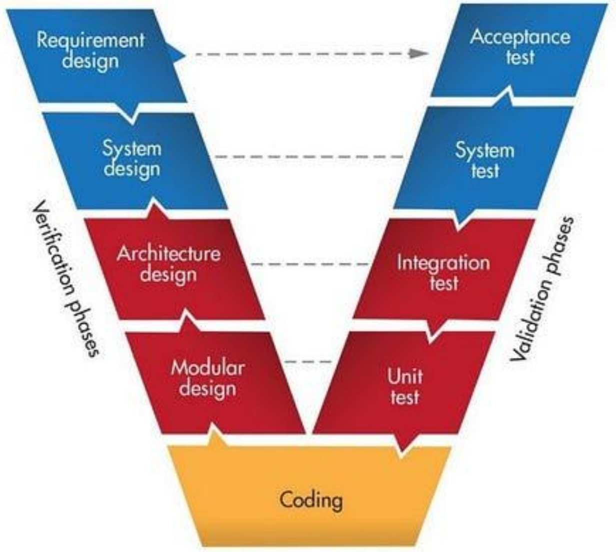
Concept → Feasibility → Specification → Development & Design Verification → Product Verification (Pilot Production 1) → Production Verification (Pilot Production 2) → Production Ramp-up & Review → Mass Production & Sales

Healthcare – Product lifecycle

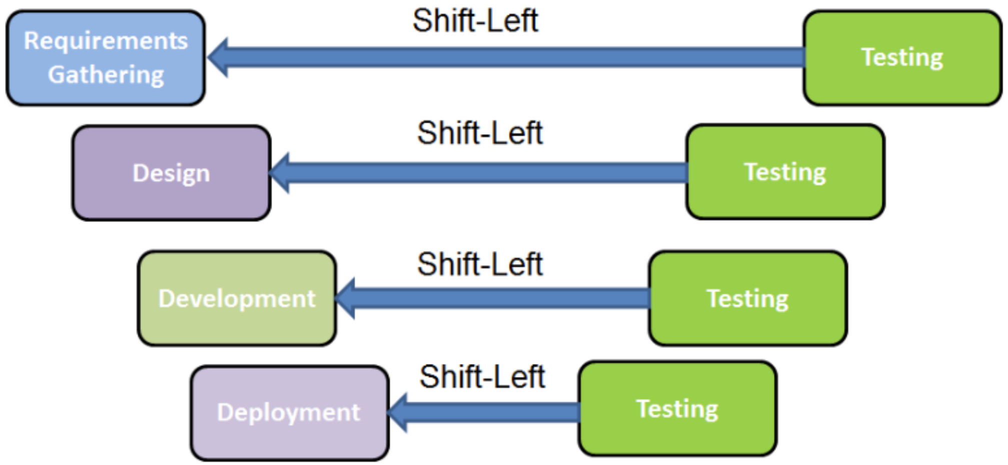


Too late!

Shift Testing Left

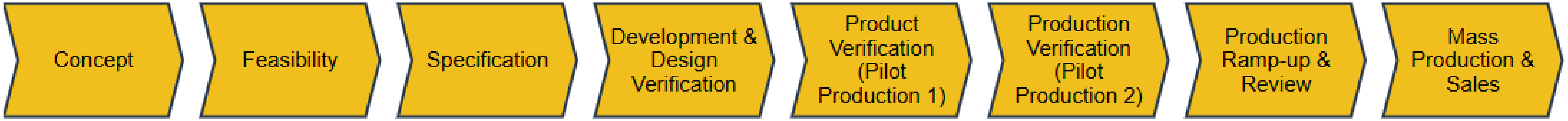


- 🕑 Earlier defect detection and prevention
- 💰 Lower cost of fixing defects
- ⚡ Faster feedback to developers
- 🔄 Improved collaboration
- ✅ Higher overall product quality



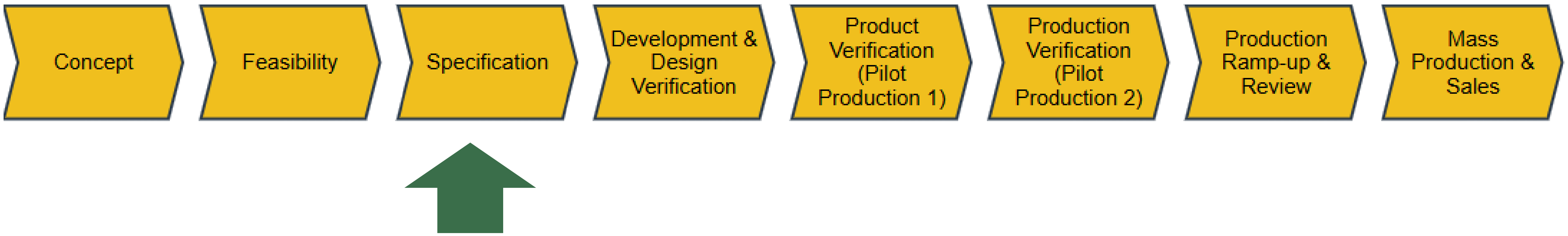
https://medium.com/qa-moments/difference-between-v-model-and-v-model-in-software-testing-2b5f4f6ff575
https://dzone.com/articles/how-shift-left-testing-can-help-your-product-quali

Healthcare – Product lifecycle



Specification Phase

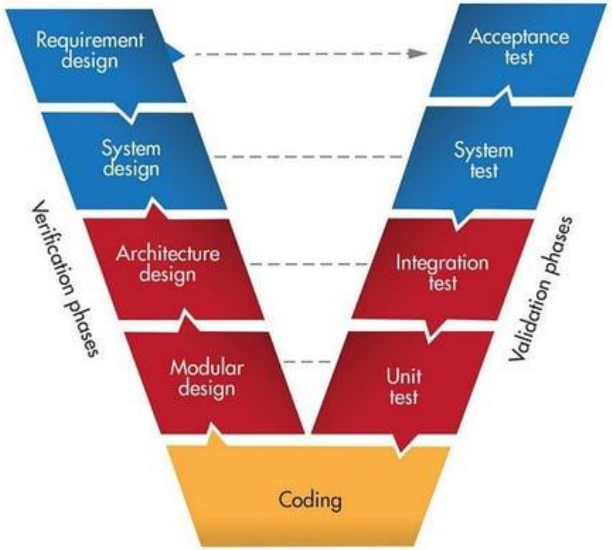
Healthcare – Specification is being written



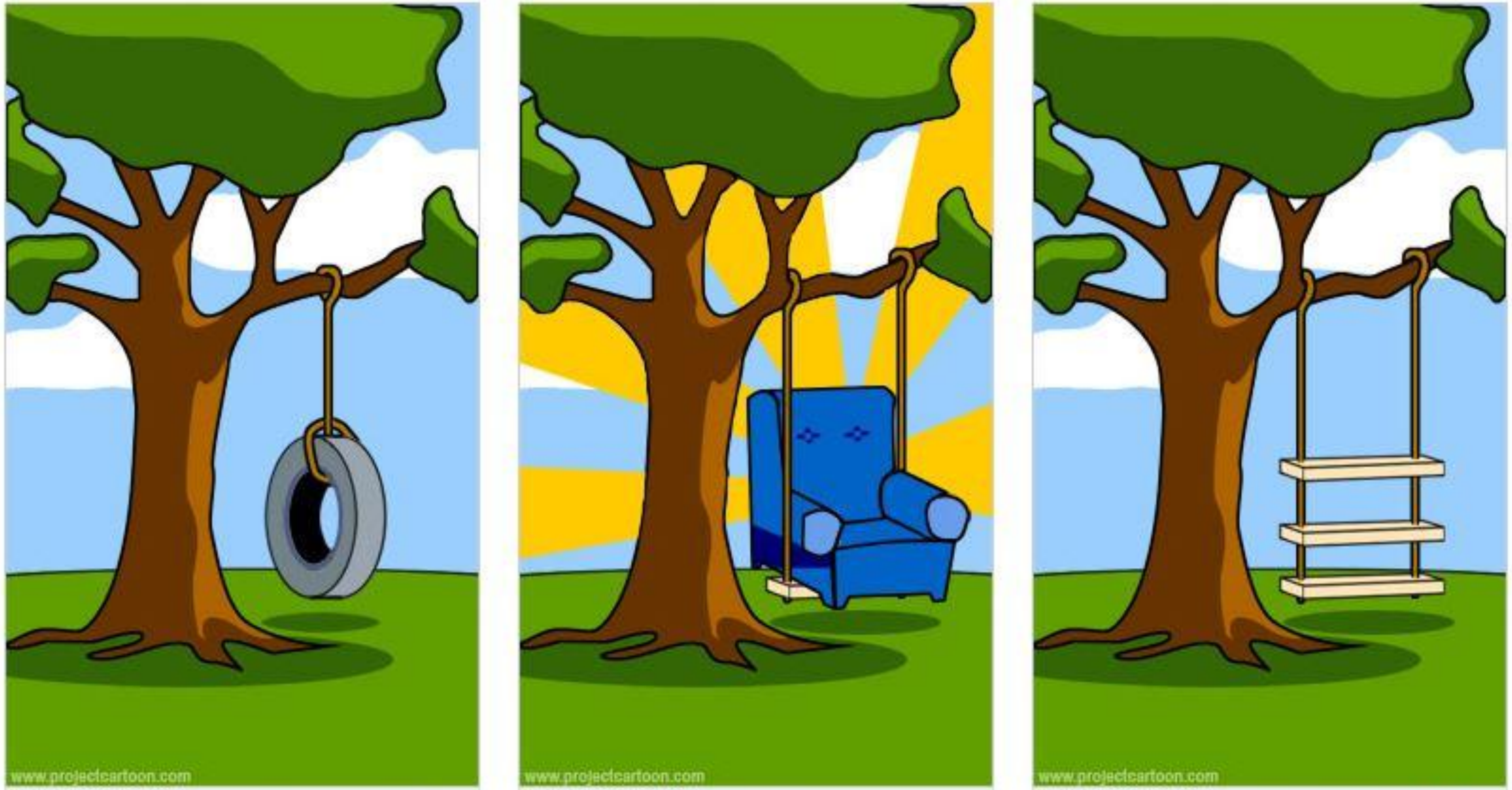

REQUIREMENTS
HD
HISTORY.COM

Customer Requirements – where we start

Customer requirements define the **desired outcomes** from the customer's point of view – the functions, features, quality levels, and constraints they expect in the final product or service.



RTX



What the business people originally had in mind

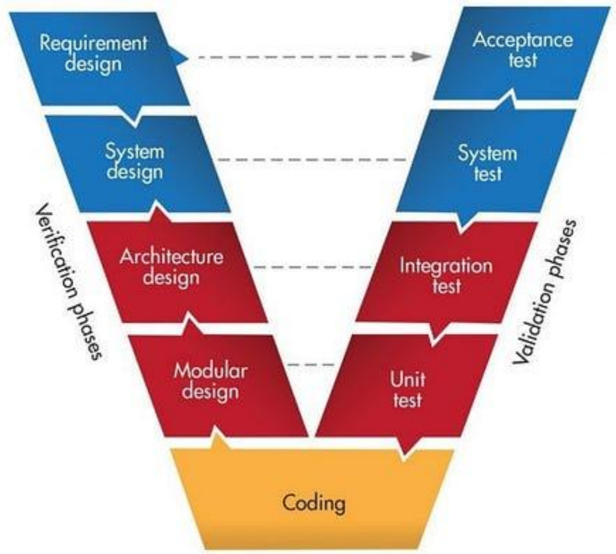
How the product was described in the specifications document

How the final product looked like

Product/Technical Requirements – where we want to end up

Technical requirements are the **detailed, measurable specifications** that describe *how* a system, product, or service will be designed, built, and operated to meet the **customer and system requirements**.

They translate **functional and performance needs** into **engineering terms**, defining the technical characteristics, interfaces, standards, and constraints necessary for implementation.



Shift left: Requirements reviewing and static analysis

The **progressive development and clarification of system or project requirements** from their initial, high-level statements into well-defined, validated, and approved specifications that can guide design, implementation, and testing.

Shift left: Requirements reviewing and static analysis

Team members reviewing and commenting on requirements

Team members agreeing on the requirements

Some success criteria (i.e Gateway goals):

- ✅ Requirements must be clear and unambiguous
- ✅ Requirements must have have clear accept criteria
- ✅ Requirements must be tracable i.e., from a customer req to a product req.
- ✅ Requirements must provide enough context so that the above is clear
- ✅ Requirements must be accepted

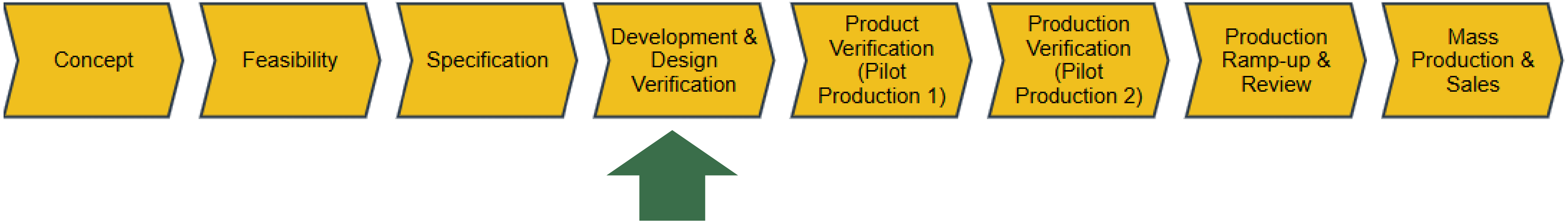
The verification and validation plan

Product Verification and Validation Plan

for

Software and Hardware Implementation

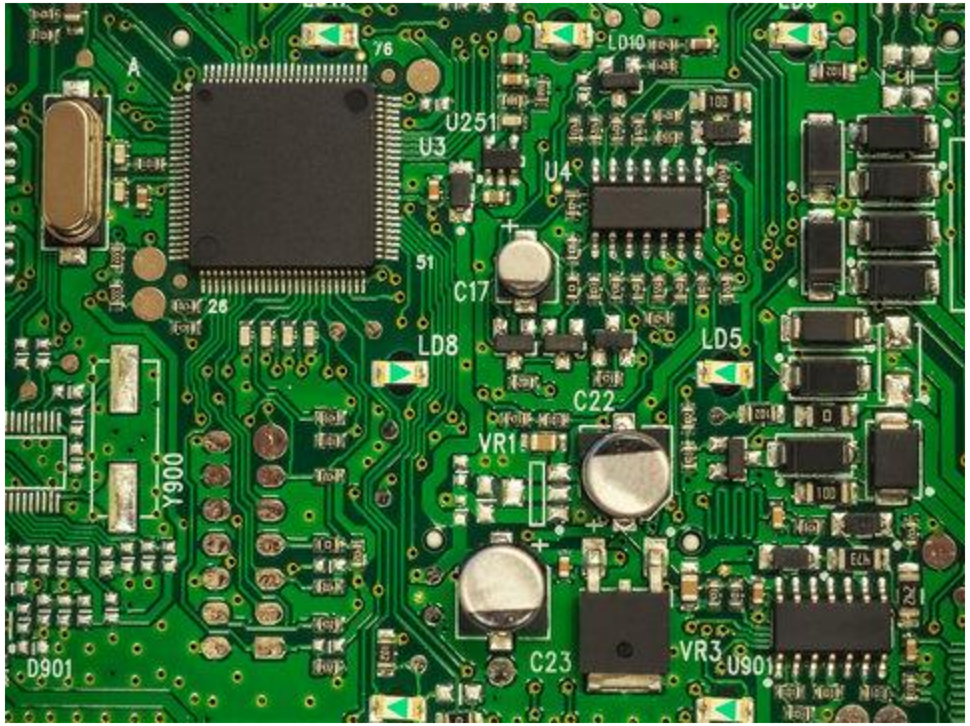
Healthcare – Development



At this point

Requirements are in an accepted state that developers can begin implementing from.

Some sort of hardware exist that the software can run on (ES hardware)



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Shifting left – Writing manual and auto tests

Testers and developers collaborate in real time on new features – Typically devs are system agnostic

Developer demos work-in-progress to tester.

Tester develops tests early making it faster to execute system tests when a build is done

Manual and automatic system tests – Difference

Manual testing is the process of **manually executing test cases** without using automation tools, to verify that a software application works as expected.

In manual testing, a human tester plays the role of an end user – using the application's features to identify defects, usability issues, or deviations from requirements.

Automated testing – is the process of **using software tools or scripts to execute test cases automatically**, compare actual outcomes with expected results, and report the findings.

Instead of a human tester manually performing each step, **automation tools simulate user actions and validate system behavior automatically.**

Manual and automatic system tests – Profiles and situations

Manual testing often involves skilled testers knows the system and what to look for.

Takes time and resources on testing especially if tests are repeated.

✅ Typically you want manual tests for subjective treatment of tests or if tests are only going to be run few times.

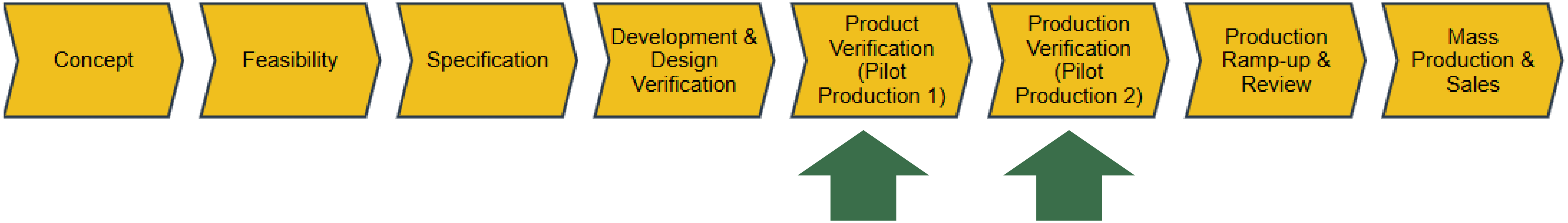
Automated testing often involves developers who knows how to write good tests.

Takes time and resources to develop and maintain tests – execution is fast.

✅ Typically you want to automate repetative and trivial tests (the ones you know if going to be run frequently).

System Testing

Healthcare – Product lifecycle



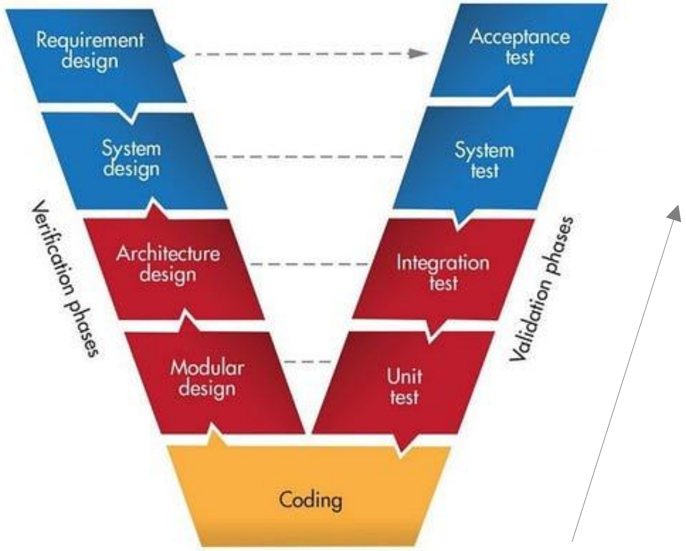
At this point

New hardware is ready (Pilot Production units, PP)

Developers are done

- Developers have a good understanding of when and where the code breaks
- Usually happens on component level i.e., unit testing and integration tests
    - I.e.,
    - Does this component have the right interface to another
    - Performance of the radiomodule,
        - Stress
        - Max capacity
        - Throughput

DOD: Developer unit and integration tests are executed
and passed before handing over to testing team



We do system tests!

When do software often break ?

- When you put it under pressure
- When you put it together with other components using other software

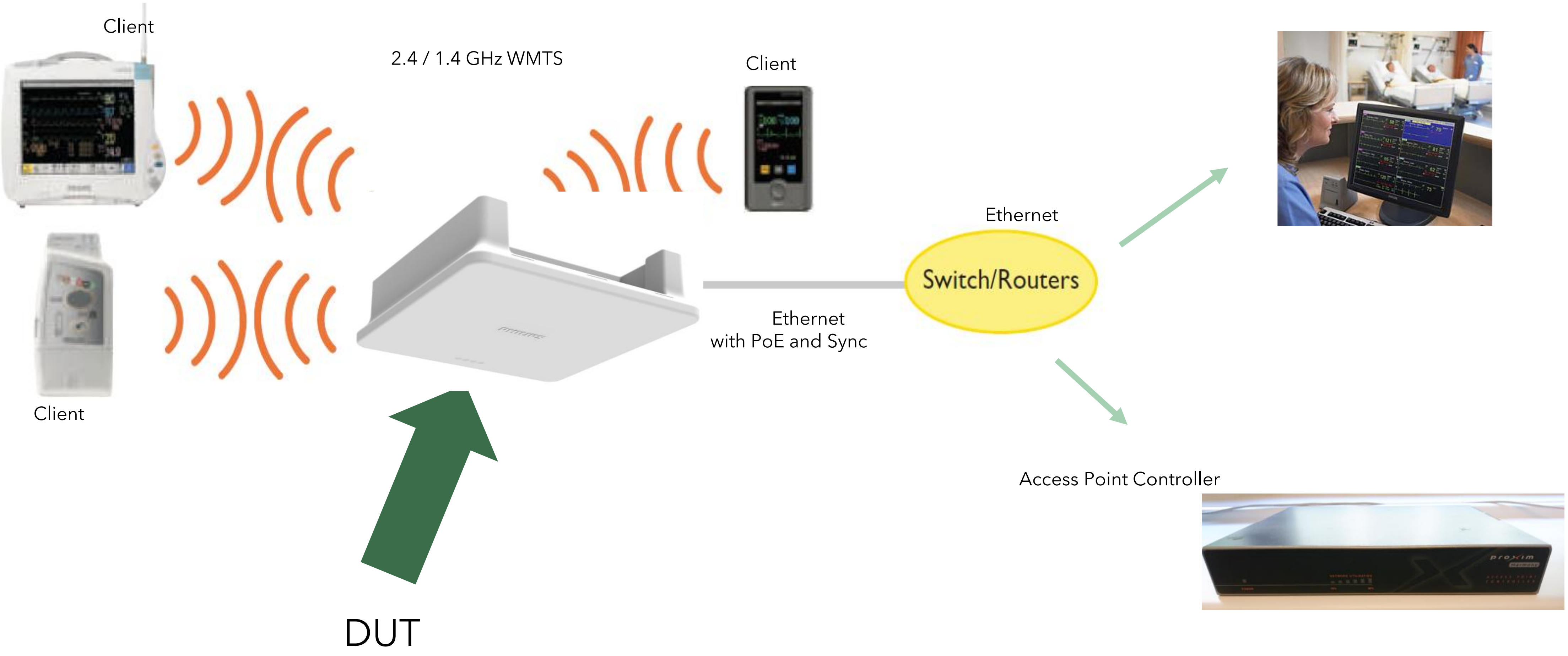
Our products does not exist in a vacuum

- They need to be compatible and operate with other products
- We need to provide system reliability – not only product reliability

Customer first!

- We are first and foremost interested in the patient and how they are impacted
- I.e., a telematry device must not lose patient monitoring

System testing mentality

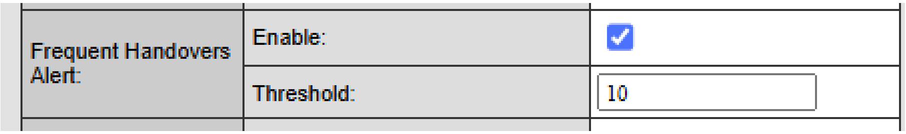


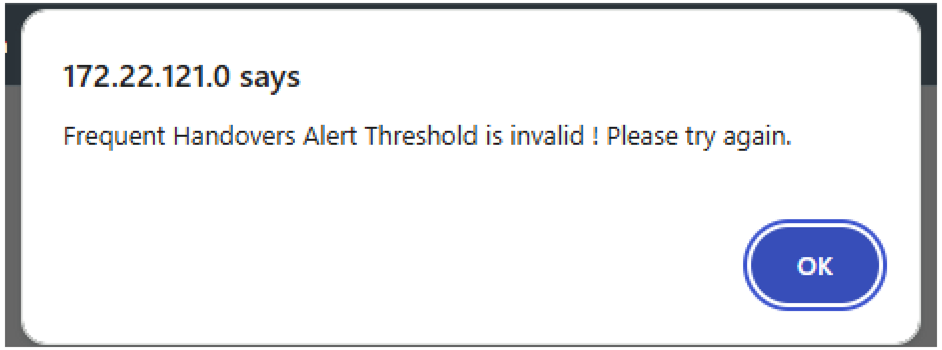
The test plan

- What is part of the software release?
- Should we do a regression?
- Should we do a full test?
- Which types of test should we do?
- What software release are we running on?
- Which hardware revision are we using?
- What other components do we have in the system and what software do they use?

Then the fun begins..

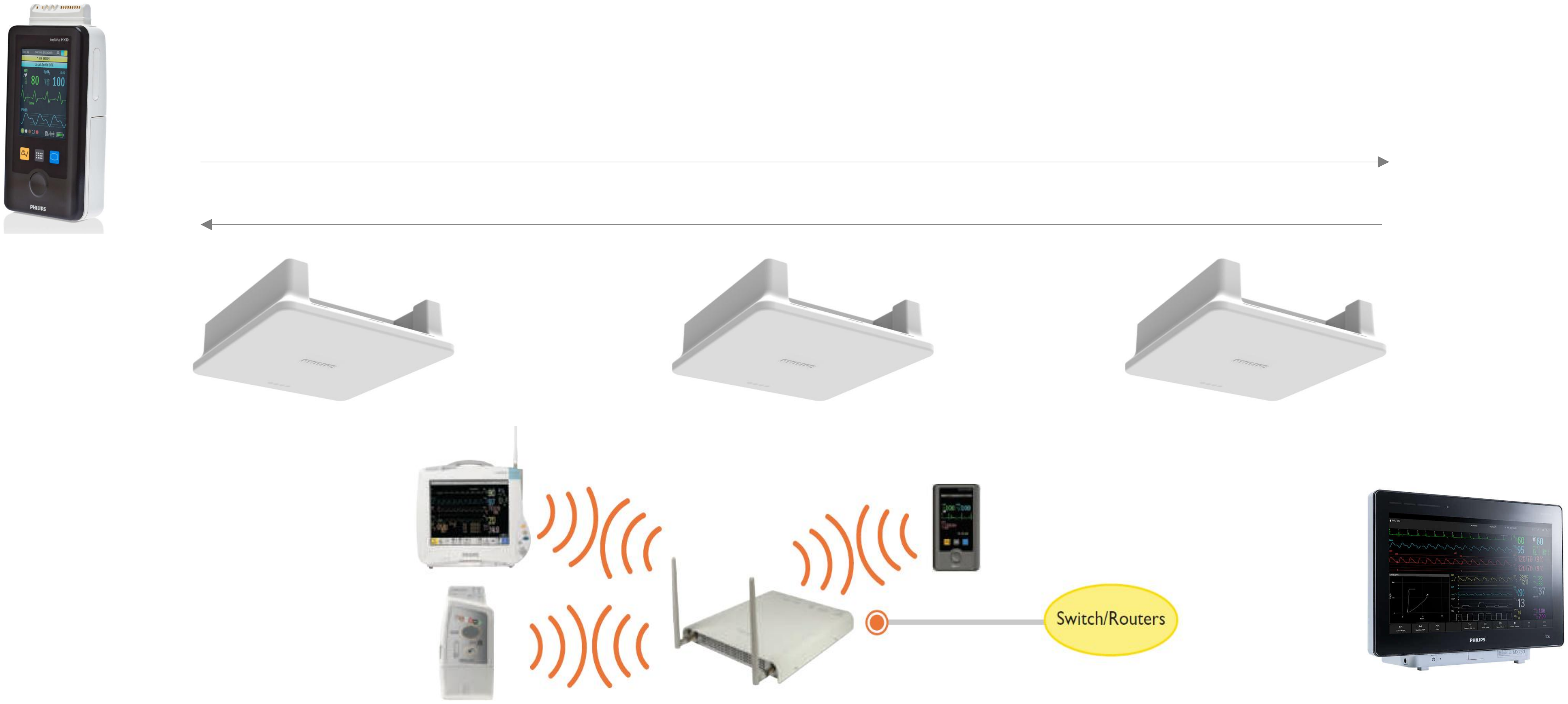
Test examples, interface validation





Test examples, roaming

Test examples, compatability



Do you take actual use into account?

Brenan Keller
@brenankeller

A QA engineer walks into a bar. Orders a beer. Orders 0 beers. Orders 99999999999 beers. Orders a lizard. Orders -1 beers. Orders a ueicbksjdhd.

First real customer walks in and asks where the bathroom is. The bar bursts into flames, killing everyone.

1:21 PM · 30 Nov 18

Do you take actual use into account?

Ad hoc testing

- Letting the testers poke the system in ways we did not expect in formal verification
- Unscripted usually

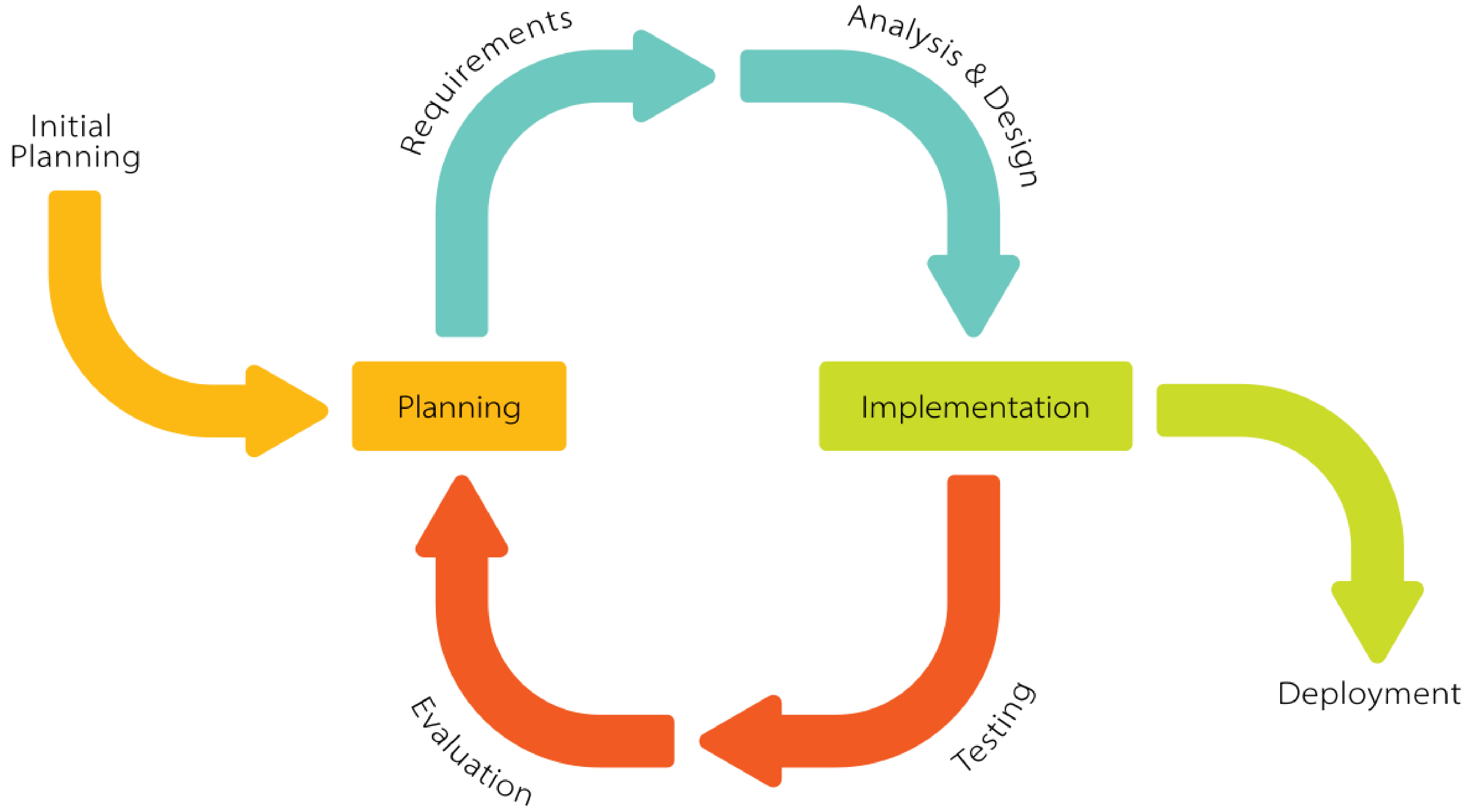
Usability testing

- Finding UI elements that may be problematic in user interaction
- Can be unscripted

Walk throughs / Testathons

- Letting field engineers walk through the system following the manual / documentation
- Finds problems we did not anticipate
- Ensures that our customers can actually use the products and figure out how to install them

What happens if the product fail the test?



Evaluate the risk

Do we go back fix it and retest everything?

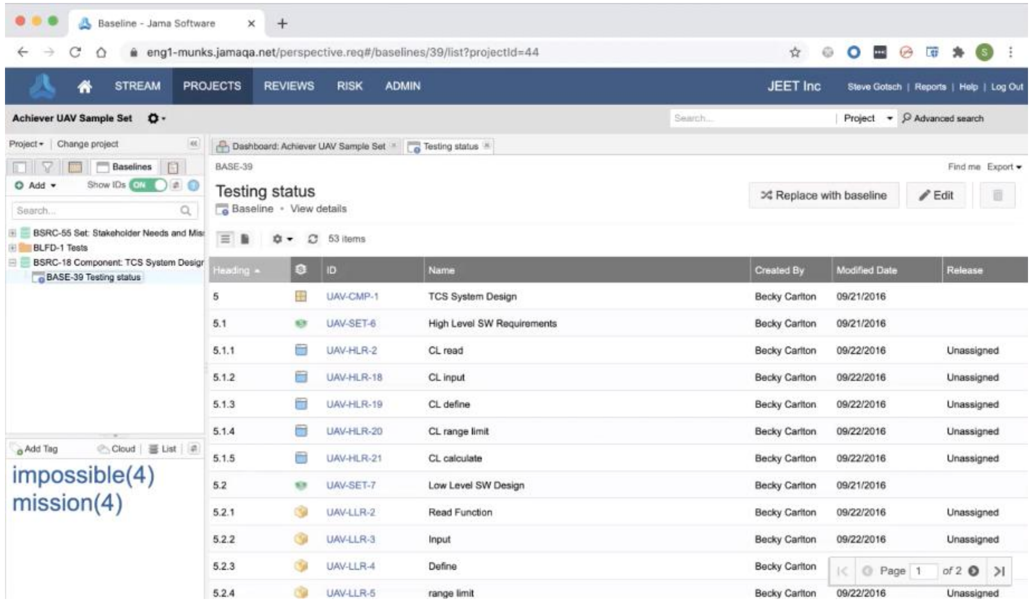
- Fx., Hardware issues, protocol.. Everything that can affect patients!
- Takes a lot of time..

Do we fix it and do a small regression?

- Cosmetic issues or something that does not affect the patients.

Either way we do retest.

Tools (Jama and Jira)



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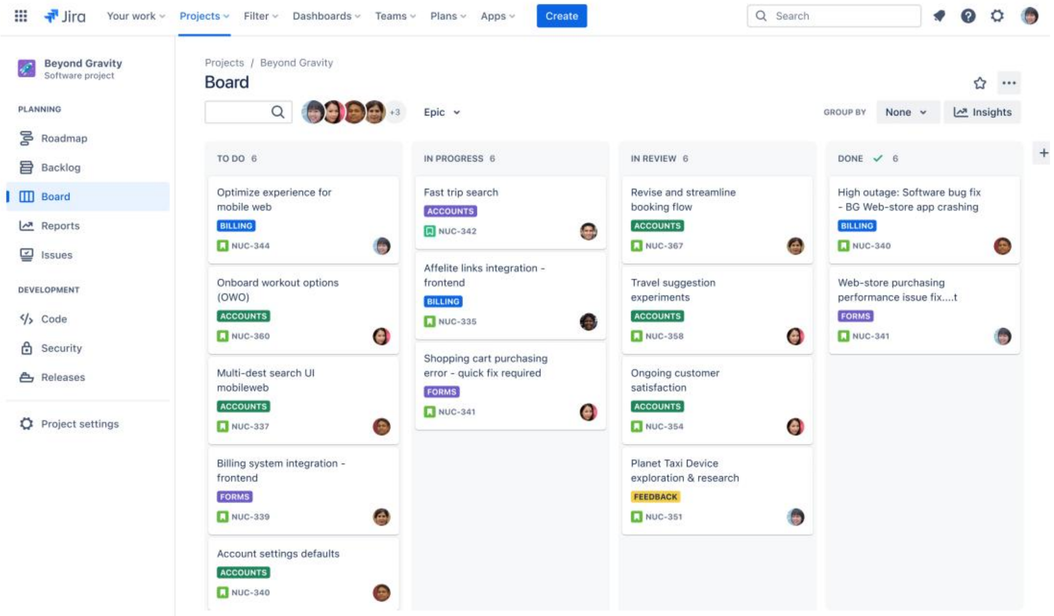


https://www.atlassian.com/agile/project-management/workflow-examples

Performance metrics

Bugs are classified based on patient impact

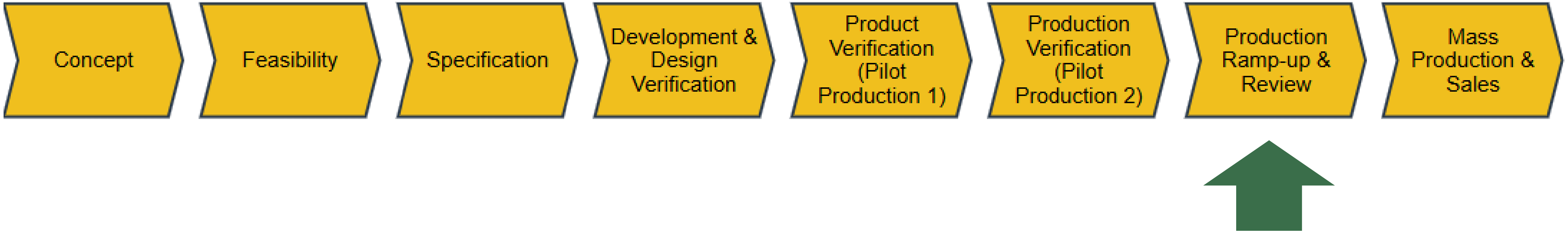
- Critical
- Serious
- Cosmetic
- Not a bug

We do not pass the product if there are any critical or serious errors

System test passed

Mass Production

Healthcare – Product lifecycle



Point of no return!

How can we make quality products?

Quality needs to be implemented from the very beginning, you cannot wait until the end and expect QA to build quality into your products magically. If QA discovers a problem in the validation phase or after, it's expensive to fix!

- We review, write, edit, and test early
- We test often on all stages in the lifecycle
- We adopts methods that will bring our product closer to the customer

Thank you!

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